

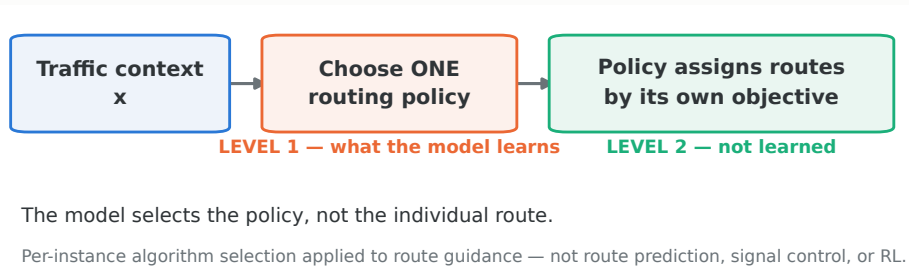
Context-Dependent Selection of Urban Routing Policies

Methodological redesign — proposal for discussion

Version 2 design. No V2 experiment has been run; all V2 quantities are specifications or hypotheses.

1 The problem

A navigation or traffic operator selects *one* routing strategy for an operating episode. Different strategies optimise different objectives, so they fail under different traffic conditions.



Can we determine, before deployment, which routing policy is appropriate for the current traffic context?

2 Why the current formulation needs redesign

1. **Weak policy complementarity.** The policies largely shared one failure mode, so the winner map reduced to a single demand variable.
2. **Highly correlated decision criteria.** Travel time and CO₂ moved together; the multi-criteria layer changed no decision.
3. **The adaptive model was close to a simple demand rule.** One split captured most of the available decision value.

The redesigned study will first test whether meaningful policy complementarity exists.

Version 1 answered its question honestly, and the answer was narrow. Complementarity is now a hypothesis under test — not a design assumption.

3 What exactly are we selecting?

Level 1 context $\longrightarrow$ routing policy *the model learns this*

Level 2 policy $\longrightarrow$ individual routes *the policy's own objective*

Provisional portfolio:

static / shortest reactive travel time
capacity-aware load balancing reliability-aware

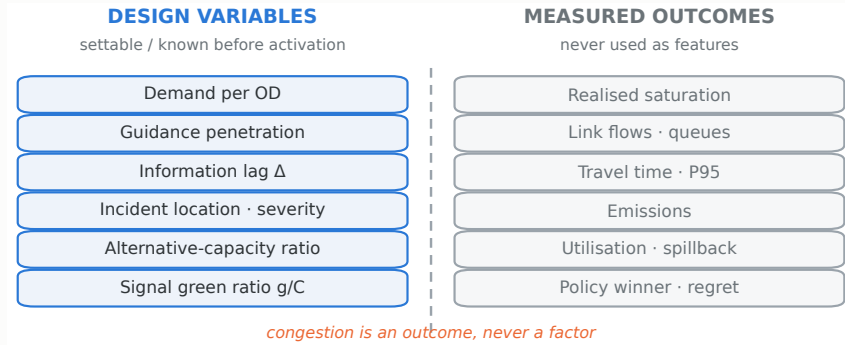
Per-instance algorithm selection applied to route guidance, with the simulator as the performance oracle. Not route prediction, not signal control, not reinforcement learning.

4 Different policies have different failure modes

Policy	Main strength	Main weakness	Triggering condition
Static / shortest	simple, efficient	overloads its own preferred route	demand approaches shortest-corridor capacity
Reactive travel time	responds to congestion	information lag, herding , oscillation	high penetration; lag long vs trip time; abrupt events
Capacity-aware	balances load across alternatives	spreads traffic that needed no spreading	light demand; one dominant path
Reliability-aware	avoids highly variable routes	detour premium when conditions are stable	stable conditions; noisy variance estimate

A portfolio is selectable only if its members fail for different reasons.

5 What creates the different regimes?



Penetration and incidents are the two factors the previous design could not vary. Without penetration there is no herding; without incidents there is no travel-time variance — so two of the four policies would have nothing to be good at.

6 Decision framework: avoiding arbitrary weights

The additive 0.5/0.5 weighting is **removed as the primary method**.

1. **Pareto screening** — remove dominated policies. No weights required.
2. **System-level mean journey time** over all traffic — ranks the survivors.
3. **Preference sensitivity** — reported only where a genuine conflict survives step 1.

If a scalar is needed: generalised cost from *published* VOT / VOR / carbon values, with a sensitivity range.

We do not invent weights merely to create a winner.

The system-level objective also charges a policy for its effect on background traffic.

7 What does the ML actually learn?

context → mechanistic features → **advantage model** → per-policy advantage → ranking → recommended policy

The model predicts **advantage relative to a baseline**

$$\Delta_a(x) = C_{\text{SBS}}(x) - C_a(x)$$

not raw travel time — centring the target on zero at the decision boundary.

Strong tabular ML (gradient-boosted trees), not a deep model: low-dimensional, interaction-heavy, context-level instances. A graph network or an RL formulation would be sophistication without a matching problem structure.

8 Robustness: when should we trust the selector?

new context → **support gate** → **confidence gate** → **ACT** or **ABSTAIN** → fixed policy

Shift types tested *separately*: interpolation · extrapolation · unseen incident · unseen OD · unseen topology.

The selector is allowed to say: I do not know.

Motivated by our own evidence: holding out a whole demand level made the previous selector *worse than the fixed policy it replaced*. Caveat on the record — conformal prediction guarantees coverage only under exchangeability, so it is a calibrated gate in-distribution and a heuristic outside it. The *support* gate is what addresses extrapolation.

9 Evaluation: does ML really add decision value?

best fixed $\rightarrow$ **mechanistic rule** $\rightarrow$ standard ML $\rightarrow$ GBDT $\rightarrow$ selective GBDT

reference: cross-fitted hindsight best policy — retrospective, not deployable

Primary metrics: regret $\cdot$ closed VBS–SBS gap $\cdot$ CVaR₉₀ regret $\cdot$ OOD risk $\cdot$ non-inferiority.

Critical test: does the model beat the strong mechanistic rule?

Beating the best fixed policy is too low a bar. If the rule wins, the finding is that the decision is mechanistically simple — and we report that.

10 Expected contribution

Hypotheses, not results:

1. Routing policies will show complementary strengths across operating regimes.
2. The strongest policy will depend on traffic, network *and information* conditions — not demand alone.
3. A mechanism-informed selector will capture useful policy-selection structure.
4. Uncertainty and abstention will reduce harmful decisions under distribution shift.
5. The study will identify when adaptive selection is genuinely useful and when a fixed strategy is sufficient.

We aim to characterise the operating regions of established routing policies and develop a context-aware, uncertainty-aware method for selecting among them.

Decision required: portfolio · scenario factors · decision objective · ML architecture

Appendix — full architecture

